MULTI-VARIABLE CALCULUS

PYQ- I. {VECTOR CALCULUS}

{2022 → Marks = 9}

Q.1(a) if $\nabla \times \vec{F} = 0$. Then it is called

(i) Solenoidal (ii) Rotational ~~(iii) irrotational~~ (iv) None

⇒ (iii) irrotational

Q.9.(b) Evaluate :

(i) div f and curl F, where $F = \text{grad}(x^3+y^3+z^3-3xyz)$.

(ii) If $F = (x+y+z)\hat{i} + \hat{j} - (x+y)\hat{k}$, show that $F \cdot \text{curl } F = 0$

⇒ (i)

⇒ Given;

$$F = \text{grad}(x^3+y^3+z^3-3xyz)$$

$$F = \nabla(x^3+y^3+z^3-3xyz)$$

Now,

$$F = I\frac{\partial}{\partial x}(x^3+y^3+z^3-3xyz) + J\frac{\partial}{\partial y}(x^3+y^3+z^3-3xyz) + K\frac{\partial}{\partial z}(x^3+y^3+z^3-3xyz)$$

$$\left\{\nabla P = I\frac{\partial P}{\partial x} + J\frac{\partial}{\partial y}P + K\frac{\partial}{\partial z}P\right\}$$

$$F = I(3x^2-3yz) + J(3y^2-3xz) + K(3z^2-3xy)$$

Here, $\text{div } F = \nabla \cdot F$

$$= \left(I\frac{\partial}{\partial x} + J\frac{\partial}{\partial y} + K\frac{\partial}{\partial z}\right)\cdot\left(I(3x^2-3yz) + J(3y^2-3xz) + K(3z^2-3xy)\right)$$

$$= \frac{\partial}{\partial x}(3x^2-3yz) + \frac{\partial}{\partial y}(3y^2-3xz) + \frac{\partial}{\partial z}(3z^2-3xy)$$

$$= 6x + 6y + 6z$$

$$\text{div } F = 6(x+y+z)$$

Hence, div F is $6(x+y+z)$.

Here $\text{Curl } F = \nabla \times F$

$$= \left(I\frac{\partial}{\partial x} + J\frac{\partial}{\partial y} + K\frac{\partial}{\partial z}\right) \times \left(I(3x^2-3yz) + J(3y^2-3xz) + K(3z^2-3xy)\right)$$

$$\left\{\begin{vmatrix} I & J & K \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x^2-3yz & 3y^2-3xz & 3z^2-3xy \end{vmatrix}\right\}$$ (Rough work)

$$= I\left(\frac{\partial}{\partial y}(3z^2-3xy) - \frac{\partial}{\partial z}(3y^2-3xz)\right)$$

$$- J\left(\frac{\partial}{\partial x}(3z^2-3xy) - \frac{\partial}{\partial z}(3x^2-3yz)\right)$$

$$+K\left(\frac{\partial}{\partial x}(+3y^2-3xz)-\frac{\partial}{\partial y}(3x^2-3yz)\right)$$

$$= I(-3x+3x) - J(-3y+3y)$$
$$+K(-3z+3z)$$

$$= 0$$

Hence, curl F is 0

(ii)

$\Rightarrow$ Given,

$$F = (x+y+1)\hat{i} + \hat{j} - (x+y)\hat{k}$$

To prove ~~that~~ ~~show~~ $F \cdot \text{curl} F = 0$.

proof:-

$$F = (x+y+1)\hat{i} + \hat{j} - (x+y)\hat{k} \quad \text{—(i)}$$

$$\text{curl} F = \nabla \times F$$

$$= \left(I\frac{\partial}{\partial x} + J\frac{\partial}{\partial y} + K\frac{\partial}{\partial z}\right) \times \left(I(x+y+1) + J - K(x+y)\right)$$

$$\left\{ \begin{vmatrix} I & J & K \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x+y+1 & 1 & -(x+y) \end{vmatrix} \right\}$$ (Rough work)

$$= I\left(\frac{\partial}{\partial y}(-(x+y)) - \frac{\partial}{\partial z}(1)\right) - J\left(\frac{\partial}{\partial x}(-(x+y)) - \frac{\partial}{\partial z}(x+y+1)\right) + K\left(\frac{\partial}{\partial x}(1) - \frac{\partial}{\partial y}(x+y+1)\right)$$

$= I(-1-0) - J(-1-0) + K(0-1)$

$= -I + 2J - K$

$\text{Curl}\,F = -I + 2J - K$ —— (ii)

Now,

$F \cdot \text{Curl}\,F$

from eq-(i) and (ii) we get,

$= \left((x+y+1)I + J - K(x+y)\right) \cdot (-I + 2J - K)$

$= -(x+y+1) + 2 + x + y$

$= -x - y - 1 + 1 + x + y$

$F \cdot \text{curl}\,F = 0$

Hence proved.

expected Ques-Ans [2023-24]

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Q.1. The gradient $f(x,y,z) = \sqrt{x^2+y^2+z^2}$ at $(1,1,1)$ is;

⇒ Given;

$$f(x,y,z) = \sqrt{x^2+y^2+z^2}$$

$$\nabla f(x,y,z) \text{ at } (1,1,1) = ?$$

$$\nabla f(x,y,z) = I\frac{\partial}{\partial x}\left(\sqrt{x^2+y^2+z^2}\right) + J\frac{\partial}{\partial y}\left(\sqrt{x^2+y^2+z^2}\right) + K\frac{\partial}{\partial z}\left(\sqrt{x^2+y^2+z^2}\right)$$

$$= I\left(\frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2x\right) + J\left(\frac{2y}{2\sqrt{x^2+y^2+z^2}}\right) + K\left(\frac{2z}{2\sqrt{x^2+y^2+z^2}}\right)$$

$$= I\frac{x}{\sqrt{x^2+y^2+z^2}} + J\frac{y}{\sqrt{x^2+y^2+z^2}} + K\frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$\nabla f(x,y,z)\ (1,1,1)$$

$$= I \cdot \frac{1}{\sqrt{1^2+1^2+1^2}} + J\frac{1}{\sqrt{1^2+1^2+1^2}} + K\frac{1}{\sqrt{1^2+1^2+1^2}}$$

$$= \frac{1}{\sqrt{3}}(I+J+K) \text{ or } \frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$$

Q.2. The gradient of the function $f(x,y,z) = \sin(xyz)$ at $(1,-1,\pi)$ is

$\Rightarrow$ Given;

$$f(x,y,z) = \sin(xyz)$$

$$\nabla f(x,y,z) \text{ at } (1,-1,\pi) = ?$$

$$\nabla f(x,y,z)$$

$$= I\frac{\partial}{\partial x}(\sin(xyz)) + J\frac{\partial}{\partial y}(\sin(xyz)) + K\frac{\partial}{\partial z}(\sin(xyz))$$

$$= I(\cos(xyz)\cdot yz) + J(\cos(xyz)\cdot xz) + K(\cos(xyz)\cdot xy)$$

$$\nabla f(x,y,z) = I(yz\cos(xyz)) + J(xz\cos(xyz)) + K(xy\cdot\cos(xyz))$$

$$\nabla f(x,y,z)(1,-1,\pi)$$

$$= I(-1\times\pi\cos(1\cdot -1\cdot\pi) + J(1\cdot\pi\cos(1\cdot -1\cdot\pi)) + K(1\cdot -1\cos(1\cdot -1\cdot\pi))$$

$$= I(-\pi\cdot(-1)) + J(\pi\cdot(-1)) + K(-1\cdot(-1))$$

$$= I(\pi) - J(\pi) + K$$

or

$$\pi\hat{i} - \pi\hat{j} + \hat{k}$$

Q.2 If $\phi(x,y,z) = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$ then $(\nabla \times \phi)$ at the point $(1,-1,1)$ is ;

=> Given ;

$$\phi(x,y,z) = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$$

$$\nabla \times \phi = ?$$

$$= \left(I\frac{\partial}{\partial x} + J\frac{\partial}{\partial y} + K\frac{\partial}{\partial z}\right) \times \left(I\,xz^3 - J\,2x^2yz + K\,2yz^4\right)$$

$$\left\{ \begin{vmatrix} I & J & K \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix} \right\} \text{(Rough work)}$$

$$= I\left(\frac{\partial}{\partial y}(2yz^4) - \frac{\partial}{\partial z}(-2x^2yz)\right)$$

$$- J\left(\frac{\partial}{\partial x}(2yz^4) - \frac{\partial}{\partial z}(xz^3)\right) +$$

$$K\left(\frac{\partial}{\partial x}(-2x^2yz) - \frac{\partial}{\partial y}(xz^3)\right)$$

$$= I(2z^4 + 2x^2y) - J(0 - 3xz^2) +$$

$$K(-4xyz - 0)$$

$$= I(2x^2y + 2z^4) + J(3xz^2) - K(4xyz)$$

$\nabla \times \phi$ at $(1,-1,1)$

$= I\left(2(1)^2(1) + 2(1)^4\right) + J\left(3(1)(1)\right)$

$- K\left(4(1)(-1)(1)\right)$

$= I(0) + 3J + 4K$

$= 3J + 4K$

or

$3\hat{j} + 4\hat{k}$.

Q.4. If $\phi(x,y,z) = 3x^2y - y^3z^2$, then $\nabla\phi$ at the point $(1,1,-2)$ is;

=> Given;

$\phi(x,y,z) = 3x^2y - y^3z^2$

$\nabla\phi$ at point $(1,1,-2)$ = ?

$\nabla\phi = I\frac{\partial}{\partial x}\left[(3x^2y) - (y^3z^2)\right] +$

$J\frac{\partial}{\partial y}(3x^2y - y^3z^2) +$

$K\frac{\partial}{\partial z}(3x^2y - y^3z^2)$

$= I(6xy) + J(3x^2 - 3y^2z^2)$
$+ K(-2y^3z)$

$\nabla\phi$ at $(1,1,-2)$

$$= I\left(6(1)(1)\right) + J\left(3(1)^2 - 3(1)^2(-2)^2\right) + K\left(-2(1)^3(-2)\right)$$

$$= 6I - 9J + 4K$$

or

$$6\hat{i} - 9\hat{j} + 4\hat{k}$$

~~Q.5. Find the directional derivative of $f(x,y,z) = xy^2 + 4xyz + z^2$ at the point $(1,2,3)$ in the direction of $3\hat{i} + 4\hat{j} - 5\hat{k}$.~~

~~⇒~~

Q.5. Find the unit normal vector to the surface $xy^2 + 2yz = 8$ at the point $(3,-2,1)$.

⇒ Given;

let $\phi = xy^2 + 2yz - 8$

$$\nabla\phi = I\frac{\partial}{\partial x}\phi + J\frac{\partial}{\partial y}\phi + K\frac{\partial}{\partial z}\phi$$

Now, $\frac{\partial\phi}{\partial x} = y^2$

$$\frac{\partial\phi}{\partial y} = 2xy + 2z$$

$$\frac{\partial\phi}{\partial z} = 2y$$

$$\nabla\phi = I(y^2) + J(2xy + 2z) + K(2y)$$

$$\nabla\phi(3,-2,1) = 4I + (-10J) - 4K$$

$$\nabla\phi(3,-2,1) = 4I - 10J - 4K$$

Unit vector normal to the surface

$$= \frac{4I - 10J - 4K}{\sqrt{16+100+16}}$$

$$= \frac{4I - 10J - 4K}{\sqrt{132}}$$

$$= \frac{1}{2\sqrt{33}}(4I - 10J - 4K)$$

Hence unit vector normal to the surface

$$= \frac{1}{2\sqrt{33}}(4I - 10J - 4K).$$

Q6 Find the directional derivative of $f(x,y,z) = xy^2 + 4xyz + z^2$ at the point $(1,2,3)$ in the direction of $3\hat{i} + 4\hat{j} - 5\hat{k}$.

→ Given;

$$f(x,y,z) = xy^2 + 4xyz + z^2$$

$$\nabla f(x,y,z) = I\frac{\partial}{\partial x}f(x,y,z) + J\frac{\partial}{\partial y}f(x,y,z) + K\frac{\partial}{\partial z}f(x,y,z)$$

Now, $\frac{\partial}{\partial x}f(x,y,z) = y^2 + 4yz$

$$\frac{\partial}{\partial y}f(x,y,z) = 2xy + 4xz$$

$$\frac{\partial}{\partial z}f(x,y,z) = 4xy + 2z$$

4 + 12 14 8-16

40 39

$$\nabla f(x,y,z) = I(y^2+4yz) + J(2xy+4xz) + K(4xy+2z)$$

$\nabla f(x,y,z)$ at point $(1,2,3)$

$$\nabla f(x,y,z)_{(1,2,3)} = 28I + 16J + 14K$$

Unit vector of direction vector (let u)

$$u = \frac{3I+4J-5K}{\sqrt{9+16+25}}$$

$$u = \frac{1}{5\sqrt{2}}(3I+4J-5K)$$

Now,

Directional Derivative $= [\nabla f(x,y,z)] \cdot (u)$

$$= (28I+16J+14K) \cdot \left[\frac{1}{5\sqrt{2}}(3I+4J-5K)\right]$$

$$= \frac{1}{5\sqrt{2}}(84+64-70)$$

$$= \frac{78}{5\sqrt{2}} = \frac{39\sqrt{2}}{5}.$$

$\phi = \phi = \phi = \phi = \phi = \phi$

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Q.7. Evaluate $A \times (\nabla \phi)$, where $A = yz^2 i - 3xz^2 j + 2xyzk$ and $\phi = xyz$.

⇒ Given;

$$A = yz^2 i - 3xz^2 j + 2xyzk$$
$$\phi = xyz$$

Now, $\nabla \phi = I \frac{\partial}{\partial x}\phi + J \frac{\partial}{\partial y}\phi + K \frac{\partial}{\partial z}\phi$

$$\frac{\partial}{\partial x}\phi = yz$$
$$\frac{\partial}{\partial y}\phi = xz$$
$$\frac{\partial}{\partial z}\phi = xy$$

$$\nabla \phi = I(yz) + J(xz) + K(xy)$$

Here, $A \times (\nabla \phi)$

$= (yz^2 I - 3xz^2 J + 2xyz K) \times (yz I + xz J + xy K)$

$= I[3x^2yz^2 - 2x^2yz^2]$
$- J(xy^2z^2 - 2xy^2z^2)$
$+ K[xyz^3 - 3xyz^3]$

$= x^2yz^2 I + xy^2z^2 J$
$- 2xyz^3 K$

Rough

$$\begin{vmatrix} I & J & K \\ yz^2 & 3xz^2 & 2xyz \\ yz & xz & xy \end{vmatrix}$$

$I[3x^2yz^2 - 2x^2yz^2] - J[xy^2z^2 - 2xy^2z^2] + K[xyz^3 - 3xyz^3]$

4 + 4
-2 -8

Q. 8. Find the directional derivative of $\phi(x,y,z) = x^2yz + 4xz^2$ at $(1,-2,-1)$ in the direction $2i - j - 2k$.

$\Rightarrow$ Given;

$$\phi(x,y,z) = x^2yz + 4xz^2$$

Now, $\nabla\phi(x,y,z) = I\frac{\partial\phi}{\partial x} + J\frac{\partial\phi}{\partial y} + K\frac{\partial\phi}{\partial z}$

$$\frac{\partial\phi}{\partial x} = 2xyz + 4z^2$$

$$\frac{\partial\phi}{\partial y} = x^2z$$

$$\frac{\partial\phi}{\partial z} = x^2y + 8xz$$

$$\nabla\phi(x,y,z) = I(2xyz + 4z^2) + J(x^2z) + K(x^2y + 8xz)$$

$\nabla\phi(x,y,z)$ at point $(1,-2,-1)$

$$\nabla\phi(x,y,z)_{(1,-2,-1)} = I(8) + J(-1) + K(-10)$$
$$= 8I - J - 10K$$

Unit vector of direction vector (u) (say)

$$u = \frac{2I - J - 2K}{\sqrt{4+1+4}} = \frac{2I - J - 2K}{3}$$

Directional derivative

$$= \left(\nabla\phi(x,y,z)\right)\cdot(u)$$

$$= (8I - J - 10K)\cdot\left(\frac{1}{3}(2I - J - 2K)\right)$$

$$= \frac{1}{3}[16 + 1 + 20]$$

$$= \frac{37}{3}$$

Hence, directional derivative is $37/3$.

Q.9. Find (i) $\vec{A} \times (\nabla\phi)$, (ii) $(\nabla\times\vec{A})\times\vec{B}$, where

$$\vec{A} = x^2 z\, i + yz^3 j - 3xy\, k$$
$$\vec{B} = 3x\, i + 4z\, j - xy\, K \text{ and}$$
$$\phi = xy^2 z$$

⇒ (i) Given;

$$A = x^2 z\, i + yz^3 j - 3xy\, k$$
$$\phi = xy^2 z$$

Now, $\nabla\phi = I\frac{\partial\phi}{\partial x} + J\frac{\partial\phi}{\partial y} + K\frac{\partial\phi}{\partial z}$

$$\frac{\partial}{\partial x}\phi = y^2 z$$

$$\frac{\partial}{\partial y}\phi = 2xyz$$

$$\frac{\partial}{\partial z}\phi = xy^2$$

$$\nabla\phi = I(y^2z) + J(2xyz) + K(xy^2)$$

Here,

$$A \times (\nabla\phi)$$

$$= (x^2zI + yz^3J - 3xyK) \times (y^2zI + 2xyzK + xy^2K)$$

$$= I[xy^3z^3 + 6x^2y^2z] - J[x^3y^2z + 3xy^3z] + K[2x^3yz^2 - y^3z^3]$$

Rough

$$\begin{vmatrix} I & J & K \\ x^2z & yz^3 & -3xy \\ y^2z & 2xyz & xy^2 \end{vmatrix}$$

$$I[xy^3z^3 + 6x^2y^2z] - J[x^3y^2z + 3xy^3z] + K[2x^3yz^2 - y^3z^3]$$

(ii) Given;

$$A = x^2zI + yz^3J - 3xyK$$
$$B = 3xI + 4zJ - xyK$$

Now, $\nabla = I\frac{\partial}{\partial x} + J\frac{\partial}{\partial y} + K\frac{\partial}{\partial z}$

$$\nabla \times A$$

$$= \left(I\frac{\partial}{\partial x} + J\frac{\partial}{\partial y} + K\frac{\partial}{\partial z}\right) \times (x^2zI + yz^3J - 3xyK)$$

$$= I\left[\frac{\partial}{\partial y}(3xy) - \frac{\partial}{\partial z}(yz^3)\right]$$
$$- J\left[\frac{\partial}{\partial x}(3xy) - \frac{\partial}{\partial z}(x^2 z)\right]$$
$$+ K\left[\frac{\partial}{\partial x} yz^3 - \frac{\partial}{\partial y} x^2 z\right]$$

Rough

$$\begin{vmatrix} I & J & K \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 z & yz^3 & 3xy \end{vmatrix}$$

$\frac{\partial}{\partial y}[3xy] - \frac{\partial}{\partial z}(yz^3) - [\frac{\partial}{\partial x}(3xy) - \frac{\partial}{\partial z} x^2 z)$
$+ K[\frac{\partial}{\partial x} yz^3 - \frac{\partial}{\partial y} x^2 z]$

$$= I\left[3x - 3yz^2\right]$$
$$- J\left[3y - x^2\right] + K\left[0\right]$$

$$(\nabla \times A) = I\left[3x - 3y^2 z^2\right] - J\left(3y - x^2\right)$$

Now, $(\nabla \times A) \times B$

$$= \left[I(3x - 3y^2 z^2) - J(3y - x^2)\right] \times \left[3xI + 4zJ - xyK\right]$$

$$= I\left[(3y - x^2)(-xy)\right]$$
$$- J\left[(-xy)\left[3x - 3y^2 z^2\right]\right]$$
$$+ K\left[4z(3x - 3y^2 z^2) - 3x(3y - x^2)\right]$$

Rough.

$$\begin{vmatrix} I & J & K \\ (3x - 3y^2 z^2) & 3y - x^2 & 0 \\ 3x & 4z & -xy \end{vmatrix}$$

$I\left[(3y - x^2)(-xy)\right] - J\left[(-xy)(3x - 3y^2 z^2)\right] + K\left[4z(3x - 3y^2 z^2) - 3x(3y - x^2)\right]$

$$= I\left[-3xy^2 + x^3 y\right]$$
$$- J\left[-3x^2 y + 3xy^3 z^2\right] +$$
$$K\left[12xz - 12y^2 z^3 - 9xy + 3x^3\right]$$

Hence, this is a $(\nabla \times A) \times B$

Q.10. If $u = x+y+z$, $v = x^2+y^2+z^2$, $w = yz+zx+xy$, prove that grad u, grad v and grad w are coplanar.

$\Rightarrow$ Given;

$$u = x+y+z$$
$$v = x^2+y^2+z^2$$
$$w = yz+zx+xy$$

Now, grad $u = \nabla u$

$$\nabla u = I\frac{\partial u}{\partial x} + J\frac{\partial u}{\partial y} + K\frac{\partial u}{\partial z}$$

Here, $\frac{\partial u}{\partial x} = 1$

$$\frac{\partial u}{\partial y} = 1$$

$$\frac{\partial u}{\partial z} = 1$$

grad $v = \nabla v$

$$\nabla v = I\frac{\partial v}{\partial x} + J\frac{\partial v}{\partial y} + K\frac{\partial v}{\partial z}$$

Here, $\frac{\partial v}{\partial x} = 2x$

$$\frac{\partial v}{\partial y} = 2y$$

$$\frac{\partial v}{\partial z} = 2z$$

grad $w = \nabla w$

$$\nabla w = I\frac{\partial w}{\partial x} + J\frac{\partial w}{\partial y} + K\frac{\partial w}{\partial z}$$

Here,

$$\frac{\partial w}{\partial x} = z + y$$

$$\frac{\partial w}{\partial y} = z + x$$

$$\frac{\partial w}{\partial z} = x + y$$

Now, $\nabla u = I + J + K$

$\nabla v = 2x\,I + 2y\,J + 2z\,K$

$\nabla w = (y+z)I + (x+z)J + (x+y)K$

We know, $\nabla u, \nabla v, \nabla w$ vectors are coplanar if and only if Determinant $= 0$

ie,

$$\begin{vmatrix} 1 & 1 & 1 \\ 2x & 2y & 2z \\ y+z & x+z & x+y \end{vmatrix} = 0$$

LHS;

$$= \begin{vmatrix} 1 & 1 & 1 \\ 2x & 2y & 2z \\ y+z & x+z & x+y \end{vmatrix}$$

$$= 1\left[2xy + 2y^2 - 2zx - 2z^2\right] - 1\left[2x^2 + 2xy - 2yz - 2zx\right]$$
$$+ 1\left[2x^2 + 2xz - 2yz - 2y^2\right]$$

$$= \cancel{2xy} + \cancel{2y^2} - \cancel{2zx} - \cancel{2z^2} - \cancel{2x^2} - \cancel{2xy} + \cancel{2yz} + \cancel{2zx}$$
$$+ \cancel{2x^2} + \cancel{2xz} - \cancel{2yz} - \cancel{2y^2}$$

$$= 0 = RHS$$

Hence, proved grad u, grad v, grad w are coplanar.

Q-) 11. Find a unit vector normal to the surface $xy^3z^2 = 4$ at the point $(-1, -1, 2)$

$\Rightarrow$ Given,

let $\phi = xy^3z^2 - 4$

Now, $\nabla\phi = I\dfrac{\partial\phi}{\partial x} + J\dfrac{\partial\phi}{\partial y} + K\dfrac{\partial\phi}{\partial z}$

Here, $\dfrac{\partial\phi}{\partial x} = y^3z^2$

$\dfrac{\partial\phi}{\partial y} = 3xy^2z^2$

$\dfrac{\partial\phi}{\partial z} = 2xy^3z$

$$\nabla\phi = I(y^3z^2) + J(3xy^2z^2) + K(2xzy^3)$$

$\nabla\phi$ at the point $(-1, -1, 2)$

$\nabla\phi(-1,-1,2) = -4I - 12J + 4K$

unit normal to the surface

$$= \frac{-4I - 12J + 4K}{\sqrt{16+144+16}}$$

$$= \frac{-4I - 12J + 4K}{\sqrt{176}}$$

$$= \frac{1}{4\sqrt{11}}(-4I - 12J + 4K)$$

$$= \frac{1}{\sqrt{11}}(-I - 3J + K)$$

Q.12 Find the directional directive of $f(x,y,z) = xy^3 + yz^3$ at the point $(2,-1,1)$ in the direction of vector $I + 2J + 2K$.

⇒ Given;

$f(x,y,z) = xy^3 + yz^3$

let $v = I + 2J + 2K$

Now, $\nabla f(x,y,z) = I\frac{\partial}{\partial x}f(x,y,z) + J\frac{\partial}{\partial y}f(x,y,z) + K\frac{\partial}{\partial z}f(x,y,z)$

Here $\frac{\partial}{\partial x}f(x,y,z) = y^3$

$\frac{\partial}{\partial y}f(x,y,z) = 3xy^2 + z^3$

$p = i + 2j$, $\hat{p} = \frac{i+2j}{\sqrt{1^2+2^2}} = \frac{1}{\sqrt{5}}(i+2j)$.

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$\frac{\partial}{\partial z} f(x,y,z) = 3yz^2$

$\nabla f(x,y,z) = y^3 I + (3xy^2 + z^3) J + 3yz^2 K$

$\nabla f(x,y,z)$ at the point $(2,-1,1)$

$\nabla f(x,y,z)(2,-1,1) = -I + 7J - 3K$

directional derivative of $f(x,y,z)$ in the direction $I + 2J + 2K = \nabla f(x,y,z) \cdot \hat{v}$

$= (-I + 7J - 3K) \cdot \frac{(I + 2J + 2K)}{\sqrt{1+4+4}}$

$= \frac{-1+14-6}{3} = \frac{7}{3}$ //

Hence, directional derivative of $f(x,y,z)$ in the direction $I+2J+2K$ is $7/3$.

Q.13 Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at the point $P(1,2,3)$ in the direction of the line PQ where Q is the point $(5,0,4)$. Also calculate the magnitude of the maximum directional derivative.

⇒ Given;

$$f = x^2 - y^2 + 2z^2$$

Now, $\nabla f = I \frac{\partial f}{\partial x} + J \frac{\partial f}{\partial y} + K \frac{\partial f}{\partial z}$

Here $\frac{\partial f}{\partial x} = 2x$

$\frac{\partial f}{\partial y} = -2y$

$\frac{\partial f}{\partial z} = 4z$

$$\nabla f = 2xI - 2yJ + 4zK$$

∇f at the point $P(1,2,3)$

$$\nabla f(1,2,3) = 2I - 4J + 12K$$

Now, $\overrightarrow{PQ} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$

$= (5-1, 0-2, 4-3)$

$= (4, -2, 1)$

let, $v = 4I - 2J + K$

Directional ~~dir~~ derivative of f in the direction of $\overrightarrow{PQ}$

$\nabla f \cdot \hat{v}$

$= (2I - 4J + 12K) \cdot \dfrac{(4I - 2J + K)}{\sqrt{16+4+1}}$

$= \dfrac{8 + 8 + 12}{\sqrt{21}}$

$= \dfrac{28}{\sqrt{21}}$

$= 2/\sqrt{3}$

Hence, direction derivative $= 2/\sqrt{3}$

Now, maximum value of directional derivative

$= |\nabla f|$

$= |2I - 4J + 12K|$

$= \sqrt{4+16+144}$

$= \sqrt{164}$

$= 2\sqrt{41}$

Hence, maximum directional derivative $= 2\sqrt{41}$